The Wrong Unit of Uncertainty: Simultaneous Conformal Bands for Repeated Cross-National Surveys

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Abstract

Comparative research routinely reads a single wave-pair mean contrast as a persistent, distribution-wide trend—attaching uncertainty to the wrong unit and over-detecting change. We give a deployable instrument: a finite-sample simultaneous band over a country's whole attitude *trajectory*, with the country as the exchangeable unit and a hierarchy of claims. The band is a clustered conformal band exact at any K for the estimated trajectory—with an explicit design-noise bound for the latent one—inside a selection-free adaptive selector: the fallback branches are nested, so at survey scale the guarantee holds under *any* selection rule, and design correction is attempted only where certifiably reliable. In the ESS (rounds 9–11), net decline in parliamentary trust is certified in six of thirty countries and persistent decline in one (Greece, on a single wave pair)—not the twenty flagged marginally. On the World Values Survey (1981–2022), a marginal wave-pair reading over-counts the persistent phenomenon by 2.6–6.5×, and what survives concentrates in post-communist and Arab-Spring-aftermath states, not the consolidated West. A subtler second error—the calibration objects are estimated from complex samples—we resolve by scope: the correction is non-identified without the design-noise law and, for the deployed procedure, unreachable at survey scale, so abstention is principled.

1 Introduction

Comparative politics increasingly treats a national public as a *distribution* rather than a mean. Trust in parliament or satisfaction with democracy in a country and year is a cumulative distribution function over a response scale, and repeated cross-national surveys—the European Social Survey (ESS; Kaminska and Lynn, 2017), the World Values Survey, the AmericasBarometer (Cohen et al., 2021)—let us track how that distribution moves across waves and *transport* it: predict one country's trajectory from the others, or flag departures from a common pattern. Prediction bands and conformal methods are a natural tool here, promising finite-sample, distribution-free coverage.

Two wrong units. A claim and its calibration can each be attached to the wrong object. The *claim* "democratic trust is declining" is typically built from a single wave-pair mean contrast, then narrated as a country-level trend. Our instrument is a finite-sample simultaneous band over a country's whole trajectory, with the country as the exchangeable unit, and an ordered hierarchy of claims (pairwise → any-pair → net → persistent → Bonferroni-across-countries); the top rung—the whole scale core declining simultaneously across every pair—is *strictly stronger* than the net-trend claim the literature typically intends, and we treat the two as co-headlines with stated severity, not as substitutes. This multiplicity-honest core needs no survey design and applies to any repeated distributional claim.

The second, subtler wrong unit is the calibration: the objects conformal transport calibrates on—other countries' attitude distributions—are themselves *estimated* from complex probability samples, with weights, clustering, and stratification. Standard practice plugs these estimates in as truth, so the band, honest about variation across countries, stays blind to each "country" being a noisy estimate of a latent population function.

The two errors are genuinely distinct. Writing $F_{c,r}(t)$ for country c's latent attitude CDF at threshold t in wave r, and $\widetilde{F}_{c,r}(t)$ for its complex-survey estimate,

$$\widetilde{F}_{c,r}(t) - \mu_r(t) \;=\; \underbrace{\big(F_{c,r}(t) - \mu_r(t)\big)}_{\text{transport error}} \;+\; \underbrace{\big(\widetilde{F}_{c,r}(t) - F_{c,r}(t)\big)}_{\text{design error}}, \tag{1}$$

where μ_r is the transport center. A conformal band calibrated on the observed $\widetilde{F}$ scores treats the sum as if it were the transport error alone: when the design error is non-negligible the band is systematically too wide, yet any correction that subtracts the survey noise can make it too narrow, because the correction is itself estimated from a small number of countries.

Contributions. We make four. *First, a diagnosis and a portable instrument.* We separate the two wrong units—the claim unit (a wave-pair contrast standing in for a persistent distribution-wide trajectory) and the calibration unit (estimated distributions treated as latent, the omitted design-error layer of (1))—and give a finite-sample simultaneous band over a country's whole trajectory, with a multiplicity-honest guarantee hierarchy that corrects the first. *Second, a provably-valid deployed procedure with an unconditional guarantee.* The band we recommend is a clustered "population" conformal band (PCB), finite-sample exact at *any* K (Theorem 3), inside an adaptive selector whose validity does not depend on how it selects: the PCB and conservative branches are *nested*, and selection among nested bands anchored at an exact band is validity-free, so the four target-blind gates are an efficiency device, not a validity assumption (Theorem 5). The design-error-correcting branch carries its own miss budget, and only at the many-unit scale where it can activate at all; at cross-national-survey K the deployed guarantee is exact $1 - \alpha$ for the estimated trajectory, with an explicit, design-noise-bounded gap to the latent one (Theorem 2b). We validate the frozen procedure in three disclosed layers, ending with a fresh sealed run on six data-generating-process families its design had never seen (Section 6). *Third, two named reanalyses* (Section 8) where the instrument changes the substantive picture. For European trust (ESS rounds 9–11), the design-aware hierarchy certifies net decline in six of thirty countries and persistence in one

(Greece—on a single wave pair, a disclosure we make prominently), not the twenty flagged marginally. Globally, a wave-pair reading of the Foa–Mounk "deconsolidation" battery on the WVS/EVS (1981–2022) over-counts the persistent phenomenon by 2.6–6.5×; aggregating per-item certifications turns the correction into a discovery—the thirteen-country *certified core* concentrates in post-communist and Arab-Spring-aftermath states, with no consolidated Western democracy certifying on more than one item. *Fourth, the scope condition that justifies this deployment.* We delineate when a design-aware correction is available, so abstention is principled rather than a concession. Without the design-noise law the latent transport-error distribution is non-identified (Theorem 1); with it, the correction is still *unreachable* at cross-national scale for the deployed procedure: the measured design-to-total ratio ρ stays severalfold below the level at which deconvolution helps, and the reliability diagnostic obeys the algorithmic floor $D \geq \sqrt{2/(K-1)}$—an identity of the frozen procedure—requiring $K \geq 94$ exchangeable populations. The two largest surveys fail in *opposite* ways: the WVS clears the size barrier ($K \approx 100$) with negligible weights-only design noise, while the ESS has genuine clustered noise but $K \leq 33$. The reader should therefore not expect the deconvolution to activate empirically; its payoff regime is many-unit settings such as small areas, demonstrated in simulation.

The remainder situates the contribution among the conformal and political-trust literatures (Section 2), sets up the problem and impossibility result (Section 3), builds the safe adaptive method and its theory (Sections 4–5), validates it in simulation (Section 6), characterizes the real-data regime (Section 7), reanalyzes political trust (Section 8), and discusses scope (Section 9).

2 Related work

Four literatures intersect here; the Supplementary Material gives the extended discussion. *Conformal prediction* (Vovk et al., 2005; Lei et al., 2018; Barber et al., 2023): our selector adapts not to temporal shift (Gibbs and Candès, 2021) but to the certified magnitude of survey design noise; within the discipline, Randahl et al. (2026) calibrate bin-conditional forecast intervals, a different object from simultaneous trajectory bands. *Cluster-as-unit conformal* (Dunn et al., 2023; Lee et al., 2023) transports guarantees to an unseen cluster but treats each cluster's data as cleanly sampled (the same clustered construction appears, for a non-survey domain, in a companion paper by the present authors; reference omitted for review); our departure is that the calibration objects are *estimated* under complex designs (Binder, 1983; Lumley, 2010). *Noisy calibration*: Wieczorek (2023) puts design into test-point weighting; Einbinder et al. (2024); Sesia et al. (2025) deconvolve a *known* label-noise model; our results show that identification premise generally fails for design noise (Theorem 1) and that even a supplied noise law faces a finite-K reliability floor (Proposition 1). *Functional simultaneous bands* (Diquigiovanni et al., 2021): unlike FDR/Bonferroni corrections of marginal tests (Benjamini and Hochberg, 1995), the "persistent" object is defined and covered directly by one band—a distinction of estimand, not error rate (Gelman et al., 2012). Substantively we engage the trust-decline and deconsolidation debates (Norris, 2017; Voeten, 2017; Wuttke et al., 2022; Valgarðsson et al., 2025; Foa and Mounk, 2016, 2017; Ouattara and van der Meer, 2023; van der Meer and Janssen, 2025; Kołczyńska and Bürkner, 2026); the closest substantive competitor is Claassen (2019, 2020), whose latent-variable panels model measurement and sampling uncertainty with model-based credible intervals—intervals whose calibration is itself contested (Tai et al., 2024)—our bands trade pooling and width for finite-sample, model-free coverage at a stated claim rung, and the two are complements. The certified core's geography intersects the post-communist disillusionment literature (Mishler and Rose, 1997; Inglehart and Catterberg, 2003), which anticipated the decline-from-euphoric-baselines pattern our certification formalizes per country.

3 Setup and the two targets

The object we deploy is the exact simultaneous band of Theorem 3; the impossibility below (Theorem 1) is a scope boundary delimiting when a design-aware correction to that band is even available.

Objects. We observe K source countries, indexed $c = 1, \dots, K$, treated as exchangeable units (the "population" or cluster is the exchangeable object, not the individual respondent). Each country has a latent attitude trajectory $F_{c,r}(t)$, a CDF over a fixed grid of thresholds $t \in \{t_1, \dots, t_T\}$ and survey rounds r. We do not see $F_{c,r}$; we see a complex-survey estimate $\widetilde{F}_{c,r}(t) = F_{c,r}(t) + S_{c,r}(t)$, where $S_{c,r}$ is design (sampling) error with a design variance $v_{c,r}(t)^2$ that a stratified-PSU / replicate-weight bootstrap can estimate. Define the transport score of a country as its worst-case standardized deviation from the transport center,

$$R_c \;=\; \max_{r,t} \frac{|F_{c,r}(t) - \mu_r(t)|}{\sigma(t)}, \qquad \widetilde{R}_c \;=\; R_c + \xi_c,$$

where $\widetilde{R}_c$ is the same quantity computed from the observed $\widetilde{F}$ and ξ_c is the induced design contamination. (Because R is a supremum, ξ_c is not exactly mean-zero or independent of R_c; the additive centered score model is a working approximation—exact at the curve scale under Theorem 4's scale family—whose failure direction, upward-biased observed scores, is conservative for the plug-in band.) The governing scalar is the design-to-*total* SD ratio

$$\rho \;=\; \frac{\mathrm{SD}(\xi)}{\mathrm{SD}(\widetilde{R})} \;=\; \sqrt{\frac{\overline{v^2}}{s_R^2 + \overline{v^2}}} \;\in\; [0, 1),$$

$\rho \to 0$ the design-noise-negligible regime, $\rho \to 1$ noise-dominated. (The simulations' generative dial is the unbounded $\mathrm{SD}(\xi)/\mathrm{SD}(R)$; all reported ρ values and the $\sqrt{1-\rho^2}$ width law use the bounded definition.)

Target and validity. We want a simultaneous band $\widehat{B}(r,t)$ that covers a *new* country's *latent* trajectory, $\Pr\left(F_{\text{new},r}(t) \in \widehat{B}(r,t)\ \forall r,t\right) \geq 1-\alpha$. Two facts frame everything. Ordinary clustered conformal on the observed scores is exact for the *survey-estimate* target $\widetilde{F}_{\text{new}}$ (it only needs exchangeability of countries); its coverage of the *latent* target differs by an explicit gap that vanishes with the design-noise scale (Theorem 2(b)—symmetry of the design law alone is *not* enough for conservativeness, so we bound the gap rather than assume its sign). Meanwhile its width is inflated by the design noise, by a factor that grows with ρ. Correcting the width requires estimating and removing the design contribution—a deconvolution.

Country exchangeability, stated honestly. The transport band's identifying assumption is a superpopulation postulate: participating countries' trajectory laws are symmetric draws—over a set that is not sampled, enters and exits rounds non-randomly, and changes designs across rounds. It is a strong idealization. Three things temper it: the headline certifications of Section 8 do not use it (they are within-country design-bootstrap statements); under approximate exchangeability the bound of Barber et al. (2023) controls the coverage gap by an unobserved total-variation distance, which we treat as qualitative and probe via a leave-one-region-out audit flagged for the archived replication; and the Bonferroni-across-countries rung protects against screening multiplicity—a distinct safeguard that does not repair non-exchangeability.

The impossibility. The central obstruction is that the deconvolution is not identified from the contaminated scores alone.

Theorem 1 (Non-identification and necessity of design information). *Let the observed score be $\widetilde{R} = R + \xi$ with $R \perp \xi$. (i) The law of R is not identified from the law of $\widetilde{R}$ without restrictions on the law of ξ: there exist distinct pairs $(\mathcal{L}_R, \mathcal{L}_\xi) \neq (\mathcal{L}'_R, \mathcal{L}'_\xi)$ with $\mathcal{L}_R * \mathcal{L}_\xi = \mathcal{L}_{R'} * \mathcal{L}_{\xi'}$. (ii) Consequently, among non-randomized bands that are measurable functions of the observed scores only, no band whose radius is almost surely smaller than the plug-in radius—the $(1-\alpha)$ conformal order statistic of $|\widetilde{R}|$—by more than the conformal discreteness slack is* *valid for the latent target on a class of design laws containing $\xi \equiv 0$. Uniform improvement over the plug-in beyond that slack therefore requires shrinking the class of admissible noise laws.*

The reading is constructive (proof in Appendix A): a band both honest and narrower than the plug-in must *supply* the design-noise law—replicate weights or a design bootstrap, exactly what Section 4 uses.

Remark 1 (Beyond surveys). Nothing here is survey-specific: the result holds whenever conformal calibration uses *estimated* objects—MRP small-area outputs, learned or imputed features, meta-analytic effects. Design/measurement information is the identifying ingredient, and its reliability at finite K is itself bounded (Proposition 1); confirmed outside surveys on a small-area calibration (Supplementary Material).

4 A nominal-safe adaptive method

Theorem 1 says the design bootstrap is the identifying input. The method turns that input into a band by choosing, target-blind, among three constructions and by never using the aggressive one unless it is provably reliable at the available number of countries.

Three branches. For calibration error curves $\{\widetilde{E}_c\}_{c=1}^K$ over threshold grid t, with per-country design SD $\widehat{v}_c(t)$ from the stratified-PSU bootstrap: *(i) Clustered PCB* takes the conformal quantile of the *unstudentized* scores $\max_t |\widetilde{E}_c(t)|$ (the U0 construction of Theorem 3); it is finite-sample exact for the survey-estimate target at any K, and its latent-target gap is the explicit two-term bound of Theorem 2(b), vanishing with the design-noise scale. *(ii) Deconvolution* studentizes by the finite-K-safe target scale $\widehat{s}^2_T(t) = \max\big(s^2_{\text{plug}}(t) - [\,\overline{\widehat{v}^2}(t) - z\ \text{SE}\,]_+,\ \text{floor}\big)$, which subtracts a *lower* confidence bound on the design variance so the correction cannot over-shrink; its width scales as $\sqrt{1-\rho^2}$ relative to PCB. *(iii) Conservative* inflates each unstudentized score by the design SD, $\max_t(|\widetilde{E}_c(t)| + z\,\widehat{v}_c(t))$ — a score that dominates the PCB score pointwise, so the conservative band *contains* the PCB band by construction. This nesting is load-bearing: selection between nested bands anchored at an exact band is validity-free (Theorem 5), so the selector needs no conditional-coverage argument.

The three questions. A naive selector asks *can we deconvolve* and *do we need to*; the safe selector adds the missing third question, *can we do it safely at this K?*

Safe selector (Algorithm 1). Let $\widehat{\rho}_{\mathrm{LCB}}$ be a conservative lower bound on ρ, $D = \max_t \mathrm{SE}(\widehat{s}_T^2)/\widehat{s}_T^2$ a reliability diagnostic, and $\widehat{\delta}_{\mathrm{UCB}}(D)$ a frozen upper bound on the finite-K coverage remainder. Deconvolution is used *iff all four gates pass*: (A) $\widehat{\rho}_{\mathrm{LCB}} > \rho_0$ (design noise materially large); (B) $\widehat{\delta}_{\mathrm{UCB}}(D) \leq \delta_{\max}$ (remainder within budget); (C) $\widehat{\mathrm{Gain}} - \Delta_g > g_{\min}$ (beats conservative by a margin); (D) stability. Otherwise the band is PCB when $\widehat{\rho}_{\mathrm{LCB}} \leq \rho_0$ and the conservative envelope when design noise is real but information insufficient. Every gate is a function of the calibration set only, so the selector is target-blind — though, by the nested-band argument of Theorem 5, validity would survive even an adversarial selection between the two anchor branches. The deconvolution branch, which is *not* nested, is charged its own miss budget $\alpha_{\mathrm{dec}} = \min\{\max(0.1\alpha, 3/(K+1)), \alpha/2\}$ (the anchors run at $\alpha - \alpha_{\mathrm{dec}}$), and only in the $K \geq 94$ regime where its gate can open at all; at the K of every certification sample in this paper the anchors keep the full α and the deployed guarantee is exact. All constants are frozen before validation (Section 5, Appendix B); when deconvolution is used the reported level is the observable $1 - \alpha - \widehat{\delta}_{\mathrm{UCB}}(D)$, a frozen *calibrated* bound validated in Section 6 (the theorem-level guarantee is Theorem 4's $\varepsilon_{K,B}$).

Deployment. The method ships as one call, `dapcb(cal_errors, v_cal, center, alpha)`, in an installable Python package (`pcb`), returning the band with the selected branch, all diagnostics, the coverage level, and the fallback reason—the user sees why the band was built as it was and reads off its finite-K guarantee.

Algorithm 1: Nominal-safe adaptive design-aware band

Input: calibration curves $\{\widetilde{R}_c\}$ (unit = country), design replicates $\{\widehat{v}_c\}$, target curve $\widehat{F}_{\text{new}}$, level α; frozen $\rho_0, \delta_{\max}, g_{\min}, \Delta_g$ and the K-only budget rule $\alpha_{\text{dec}}(K)$

set $(\alpha_{\text{anchor}}, \alpha_{\text{dec}})$ by the frozen budget rule (full α to the anchors when gate B is infeasible at this K);

compute diagnostics s, $\widehat{s}_T$, $\widehat{\rho}_{\text{LCB}}, D, \widehat{\delta}_{\text{UCB}}(D)$; the nested unstudentized anchor radii $r_{\text{PCB}}, r_{\text{con}}$ at level α_{anchor}; and r_{dec} at level α_{dec};

if $\widehat{\rho}_{LCB} \le \rho_0$ **then**
  branch $\leftarrow$ PCB;
else if $\widehat{\delta}_{UCB}(D) \le \delta_{\max}$ ***and*** *stable* ***and*** $\widehat{Gain} - \Delta_g > g_{\min}$ **then**
  branch $\leftarrow$ deconvolution;
else
  branch $\leftarrow$ conservative;

return $\widehat{F}_{\text{new}} \pm r_{\text{branch}}$, branch, and diagnostics;

5 Theory

We state validity for the oracle, the estimated-law procedure, and the deployed selector, then efficiency. Proofs are in Appendix A; each theorem has a machine-checked contract test.

Theorem 2 (Oracle validity)**.** *Suppose the countries are exchangeable and the design law is known.* (a) *The clustered band is finite-sample valid for the survey-estimate target:* $\Pr(\widetilde{F}_{new} \in \widehat{B}) \ge 1 - \alpha$, *using exchangeability only.* (b) *For the latent target the same band satisfies, for every* $u > 0$, $\Pr(F_{new} \in \widehat{B}) \ge 1 - \alpha - \Pr(\xi < -u) - \Pr(\widetilde{R} \in (\widehat{q} - u, \widehat{q}])$*: a noise-tail term plus a local anti-concentration term, vanishing with the design-noise scale ($\rho^{2/3}$ under variance-only tails; ρ up to logarithms under sub-Gaussian noise; Appendix A states the conditions, including where independence is used). Symmetric design noise alone does* not *make the band conservative for the latent target—a bimodal latent law furnishes a counterexample (Supplementary Material)—which is why we report the observed ρ rather than assuming conservativeness.* (c) *The deconvolution band with the true target scale is valid for the latent target—finite-sample* exact *under the scale-family assumption (A1) of Theorem 4—with width a factor $\sqrt{1-\rho^2} + o(1)$ of the clustered band.*

Theorem 3 (Exact finite-K validity of the deployed base band)**.** *Let the base band use* *the* unstudentized *cluster score* $R_c = \max_t |E_c(t)|$ *(no data-dependent modulation), with a center satisfying the symmetric-construction condition: fixed, split-fold, per-unit from that unit's own data (the deployed own-previous-round centers), or a symmetric function of all* $K{+}1$ *curves (Appendix A states the four cases). If the countries are exchangeable, then for* every K *the band* $\widehat{B} = \mathit{center} \pm \widehat{q}$*, with* $\widehat{q}$ *the* $\lceil(1-\alpha)(K{+}1)\rceil$*-th order statistic of* $\{R_c\}$*, satisfies* $\Pr(\widetilde{F}_{new} \in \widehat{B}) \geq \lceil(1-\alpha)(K{+}1)\rceil/(K{+}1) \geq 1-\alpha$*, distribution-free—exact for the* survey-estimate *trajectory (T1); the latent trajectory (T2) adds the explicit gap of Theorem 2(b), bounded through the reported* ρ*. Two asymmetries are excluded by construction: in-sample modulation (an* $O(g/K)$ *self-inclusion deficit) and grand-mean centers containing their own unit (an anti-conservative* $O(1/K)$ *bias, removed by leave-one-out centers); Appendix A quantifies both. On the near-homoskedastic real curves the width cost of dropping the modulation is* $\leq 5\%$*.*

Theorem 4 (Estimated-law validity)**.** *Assume the observed and latent score curves sit in a common scale family indexed by the design variance (A1), the sup-score law satisfies an anti-concentration bound near its upper quantile (A2), and fourth moments and an unbiased size-B design bootstrap exist (A3) — Appendix A states all three precisely. Then* (i) *with oracle scales the deconvolution band is finite-sample* exact *for the latent target at every* K*, and* (ii) *with estimated scales it satisfies* $\Pr(F_{new} \in \widehat{B}) \geq 1 - \alpha - \varepsilon_{K,B}$ *with* $\varepsilon_{K,B} = O\big(\sqrt{\log(KT)/\min(K,B)}\big)$*, proved by a sandwich and concentration argument with constants depending only on the constants of (A1)–(A3). The finite-*K*-safe scale* $\widehat{s}^2_T$ *(Section 4) subtracts a lower confidence bound on the design variance, making the scale-error term one-sided (wide, not narrow) outside an* α_2*-probability event. Some shape restriction is not removable—the bimodal counterexample of Theorem 2(b) defeats any latent-target band without one—and (A1) is one sufficient choice, strong enough to make the distributional matching exact; alternatives (e.g. smoothness conditions under a known noise law, as in the deconvolution literature) trade finite-sample exactness for weaker assumptions.*

Theorem 5 (Safe-adaptive finite-K validity—the deployed pipeline)**.** *Deploy Algorithm 1* *with unstudentized PCB and conservative branches (scores* $\max_t |E_c(t)|$ *and* $\max_t(|E_c(t)| + z\widehat{v}_c(t))$*, both at level* α_{anchor}*) and the deconvolution branch at its own budget* α_{dec}*, where* $(\alpha_{anchor}, \alpha_{dec}) = (\alpha, 0)$ *when the reliability gate is infeasible at the observed* K *(Proposition 1:* $K < 94$*, which covers every certification sample analyzed in this paper, including the WVS* $\geq$ *2-wave sets at* $K = 59$*–77) and otherwise* $\alpha_{dec} = \min\{\max(0.1\alpha,\ 3/(K{+}1)),\ \alpha/2\}$ *with* $\alpha_{anchor} = \alpha - \alpha_{dec}$*, a frozen function of* K *alone. Then, finite-sample and with* no conditions on the selection rule,

$$\Pr\left(\widetilde{F}_{new,r}(t) \in \widehat{B}(r,t)\ \forall r, t\right) \ \geq\ 1 - \alpha - \varepsilon_{K,B}\,\mathbf{1}\{K \geq 94\},$$

where $\varepsilon_{K,B}$ *is Theorem 4's estimated-law remainder, attached to the deconvolution branch only—the* $K \geq 94$ *clause therefore inherits that theorem's assumptions (A1)–(A3), while the* $K < 94$ *clause needs exchangeability only—and reported through the observable diagnostic* $\widehat{\delta}_{UCB}(D)$ *when that branch fires. As in Theorem 3, the* $K < 94$ *statement is exact for the survey-estimate target T1; the latent target T2 adds the T1→T2 gap of Theorem 2(b). (The deconvolution branch, when it fires, targets T2 directly under (A1)–(A3).)*

The proof (Appendix A) uses no conditional-on-selection step: a nested-band lemma (the conservative score dominates the PCB score pointwise, and any selection between nested bands anchored at an exact band is validity-free—an instance of nested-conformal machinery (Gupta et al., 2022; Yang and Kuchibhotla, 2024); our contribution is engineering the branches so it applies) plus a union bound charging the non-nested deconvolution branch its own budget, only where its gate can open. The gates therefore carry no validity burden; at survey scale the deployed guarantee is exact $1-\alpha$ for the survey-estimate trajectory, full stop, and honest for the latent one through the reported, ρ-bounded gap. Two scope notes. First, in the $K \geq 94$ regime the *reported* level $1 - \alpha - \widehat{\delta}_{\mathrm{UCB}}(D)$ is a frozen calibrated bound (Supplementary Material, S3), not a theorem—the theorem-level statement is the display above, and without (A1)–(A3) the worst case when deconvolution fires is $\alpha_{\mathrm{anchor}} + \Pr(\widehat{J} = \mathrm{dec})$.

Second, Theorem 3's condition covers the deployed certification and validation modes; the unsurveyed-target mode with leave-one-out centers carries a residual $O(1/K^2)$ asymmetry bounded only heuristically (Appendix A)—users needing a strict guarantee there should use a split-fold center.

Efficiency. *(AW-1, proved)* the low-ρ deconvolution/PCB width ratio is $1 - \frac{1}{2}\rho^2 + O(\rho^4)$; *(AW-2)* conservative $\supseteq$ PCB holds pointwise, its Bonferroni domination resting on an unproved expansion; *(AW-3, proved)* selection cannot inflate miscoverage (Theorem 5). AW-4 (open) and the fitted $O(g/K)$ modulation deficit are in Appendix A.

6 Simulation and a design-preserving stress test

Simulation establishes two facts: attaching the band to the wrong unit collapses coverage, and the safe selector meets its guarantees on data it has never seen.

6.1 The wrong unit collapses coverage

On a ground-truth simulation ($K = 30$ exchangeable countries, $T = 6$ thresholds, errors correlated across rounds and thresholds; Table 1) we ask each band for *trajectory* coverage—the whole country curve-path at once. All three bands are unstudentized and finite-sample constructed, so only the unit of the score varies. The per-round band, marginally valid at the round level, covers the whole trajectory only 0.82 at $L = 2$ and 0.50 at $L = 8$; the marginal band collapses faster; the country-trajectory band holds ≈ 0.90 at every L, because the trajectory is a single exchangeable unit. This is the coverage face of the paper's thesis: the same mechanism drives the certification-count collapse of Section 8. Pre-registered, reported as produced (`e28`).

Severity: what each rung can detect. At ESS-like design noise (threshold SE 0.012; sup-t certification, $\alpha = 0.10$, four-threshold core; experiment E32), 80% power at the *net*

Table 1: Whole-trajectory coverage (4000 reps, nominal 90%) as trajectory length L grows.

L (rounds)	marginal (per point)	per-round	trajectory (ours)
2	38.2%	81.7%	**90.0%**
4	16.1%	68.2%	**90.2%**
6	7.1%	58.2%	**90.3%**
8	3.5%	49.8%	**90.9%**

rung needs a per-pair decline of 0.02–0.03 CDF points over four pairs—realistic secular erosion—while the *persistent* rung needs 0.06–0.08 per pair, crisis-scale, and its power *falls* as the trajectory lengthens. Hence non-certification at the persistent rung is weak evidence of absence; the powered instrument for secular decline is the net rung, which Section 8 co-headlines. Real-data injection is flagged for the archived replication.

Validation, in three disclosed layers. *(i)* A sealed holdout (450 cells: ten DGP families $\times$ $K \in \{25, \ldots, 320\}$ $\times$ nine ρ; 800 reps each) was frozen and run once—but an audit found its scorer evaluated the S2 studentized band rather than the deployed band, and its findings motivated the switch to U0 and the Theorem 5 architecture, so we label it development. *(ii)* A corrected-scorer rerun of the same grid scores the deployed band verbatim (fresh seed; the original sealed config was not preserved—disclosed): all 450 cells floor-compatible, worst 0.882, deconvolution active only at $K = 320$, low-ρ width 1.005$\times$ plain PCB. *(iii)* Because both layers had touched the final design, we sealed one further validation on six *new* DGP families and grids, script hash recorded before first execution, run once, reported as produced including a disclosed first-seal generator bug (E33): 120/120 cells floor-compatible, worst 0.878, zero deconvolution activations below $K = 94$ as the floor requires. Full tables and the finite-K mechanism are in the Supplementary Material.

Design-preserving stress test, honestly bounded. Subsampling the real Americas-Barometer, the deployed selector never fires; but the pooled $\widehat{\rho}$ turnover at deep subsampling coincides with ≈ 1 PSU per stratum, where the m-of-m bootstrap's variance estimate

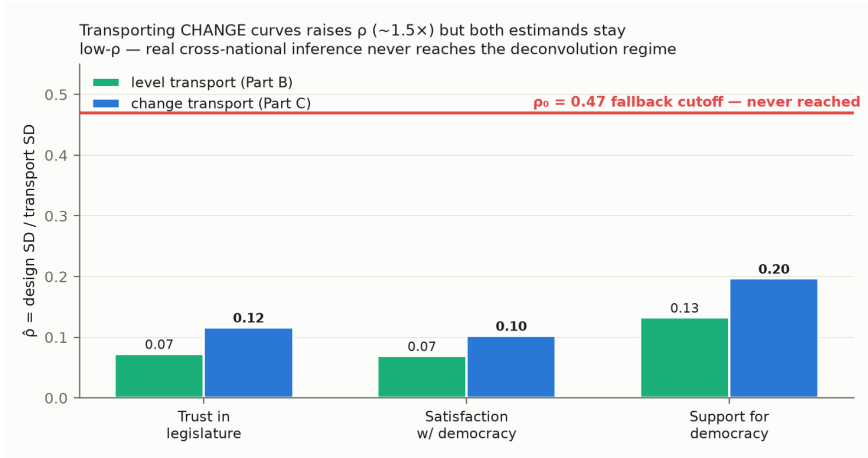

Figure 1: Design-to-total SD ratio $\widehat{\rho}$, AmericasBarometer (levels and change; $B = 200$ design bootstrap). All country-outcomes sit well below $\rho_0 = 0.47$; estimates are downward-biased floors (text).

collapses—so the sweep demonstrates a failure mode of the $\widehat{\rho}$ diagnostic under sparse-PSU designs (coverage stayed at or above nominal throughout), not a saturation law; the regime characterization of Section 7 rests on the full-sample estimates instead (Supplementary Material).

7 Evidence from the ESS and the AmericasBarometer

The regime characterization. On real cross-national data the honest band reduces to clustered conformal—and the reason is instructive. Across both surveys, both estimands (levels and wave-to-wave change), and three outcomes, national cross-national inference is a uniformly *low design-noise* regime: $\widehat{\rho} \leq 0.22$ on LAPOP (Figure 1), at most 0.29 in the ESS subgroup scan, well below the $\rho_0 = 0.47$ cutoff. Two directional caveats attach to every $\widehat{\rho}$ we report: the PSU bootstrap resamples m-of-m without Rao–Wu–Yue rescaling (Rao et al., 1992) and single-PSU strata contribute no variance, so the estimates are downward-biased *floors*; the qualitative conclusion survives because the gap to ρ_0 is severalfold. The safe selector thus reduces to clustered PCB on all real data—not by assumption but by its own low-ρ gate.

Why the deconvolution is unreachable, precisely.

Proposition 1 (Survey-scale unreachability)**.** *The deployed selector activates the deconvolution branch only if a* need *gate* $\widehat{\rho}_{\mathrm{LCB}} > \rho_0$ *and a* reliability *gate* $D \leq \tau_D$ *both hold, with* $D = \max_t \widehat{\mathrm{SE}}(\widehat{s}_T^2)/\widehat{s}_T^2$*. Two facts bound these, both* algorithmic identities of the frozen procedure *(Appendix A separates this deterministic statement from its kurtosis-dependent statistical interpretation).* (a) $\rho = \sqrt{\overline{v^2}}/s_{\mathrm{plug}} \in [0, 1)$ *with* $s_{\mathrm{plug}}^2 = s_R^2 + \overline{v^2}$*: the need gate fires only when design noise is an appreciable fraction of the between-population signal.* (b) *Since* $\widehat{s}_T \leq s_{\mathrm{plug}}$ *by construction,* $D \geq \sqrt{2/(K-1)}$*; the reliability gate requires* $K \geq 1 + 2/\tau_D^2$*, i.e.* $K \geq 94$ *at the frozen* $\tau_D = 0.147$*.*

The two requirements pull apart on real surveys (Figure 2). On the ESS, the noisiest subpopulations (youth age bands) raise $\widehat{\rho}$ only to 0.29, while $K \leq 33$ leaves $D \geq 0.25$: *neither* gate opens (42 cells scanned, zero activations). The WVS is the mirror image, with one denominator made explicit: its *full item-coverage* sets ($K = 95$–105) make the reliability gate feasible (four items pass at $D = 0.139$–0.142; importance-of-democracy misses at 0.1473), but the *certification* samples of Section 8 (≥ 2 waves, $K = 59$–77) sit below 94, where the floor closes the gate outright—and in either denominator its weights-only $\widehat{\rho}_{\mathrm{LCB}} \leq 0.09$ (itself biased low by the missing clustering) fails the need gate decisively. The survey large enough to estimate the deconvolution has almost no design noise to remove; the survey with real clustered design noise has far too few countries. The barriers transfer beyond surveys; the one many-unit setting where the need gate fires (MRP-style small areas) is demonstrated in the Supplementary Material, along with the AmericasBarometer's design-effect audit (median deff 1.13–1.20, up to 1.9).

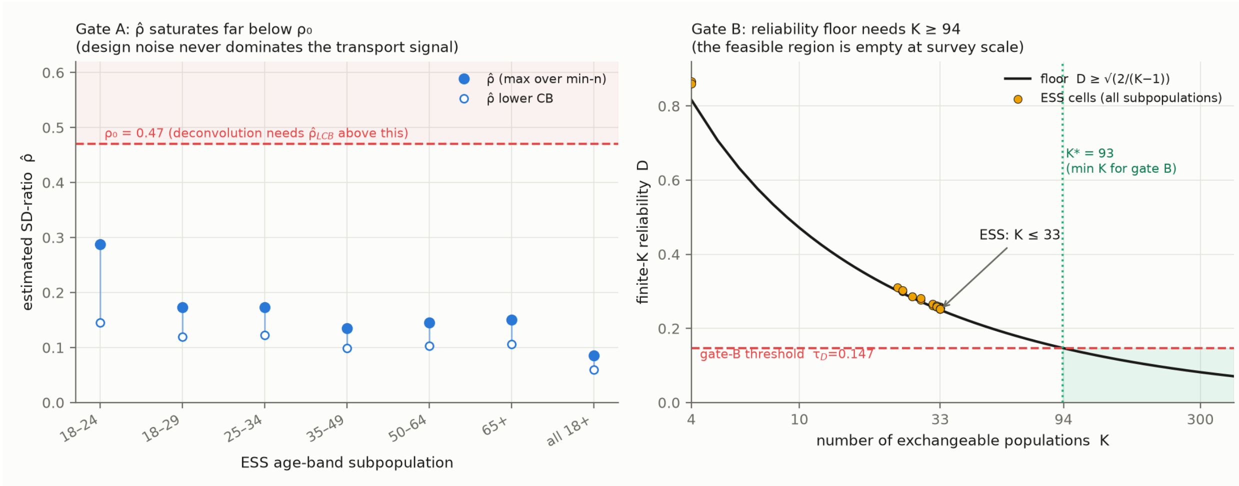

Figure 2: The two barriers on real data. *Left:* ESS age-band subpopulations (42 cells), $\widehat{\rho}$ vs $\rho_0 = 0.47$ (max 0.29). *Right:* reliability D vs its floor $\sqrt{2/(K-1)}$; the gate needs $K \geq 94$, reached only by WVS full item-coverage sets, where the need gate fails.

8 Political reanalysis: how many democracies really declined?

We analyze trust in parliament (`trstprl`) and satisfaction with democracy (`stfdem`) in ESS rounds 9–11 (fieldwork 2018–2024; the window in which PSU/stratum identifiers ship in the integrated file—earlier rounds' sample-design files exist separately and would extend the trajectory, a flagged revision), asking at each rung of the hierarchy in how many countries a decline survives.

The comparative literature has grown skeptical of a continent-wide trust decline (Norris, 2017; Voeten, 2017; Wuttke et al., 2022; Valgarðsson et al., 2025); our contribution is a formal instrument that says so, and the wrong-unit diagnosis of *why* the over-reading arises. The dominant global trend estimates (Valgarðsson et al., 2025) rest on latent-trait models whose intervals carry no design-based uncertainty layer (Tai et al., 2024) and whose data end in 2019, where our window begins. A mean can fall because a mode hollows out or a tail thickens, with different substantive meanings (DiMaggio et al., 1996); requiring the distribution's core to shift is the stronger claim.

A hierarchy of claims. The claims differ in their unit and are not interchangeable. A *pairwise* claim asks whether some pair of waves shows a distributional shift; an *any-pair* claim aggregates these within a country; a *net* claim asks whether the first-to-last span declined; a *persistent country-wide* claim asks whether the *whole* distribution declined *simultaneously* across the trajectory—a single band that must hold over all wave pairs and all thresholds at once; and a *Bonferroni-across-countries* claim further controls the number of countries examined. Each rung is strictly stronger than the last, and conflating them is exactly the over-detection this paper is about.

Which instrument certifies what. Two instruments carry different guarantees. Within-country *certification* is a design-based simultaneous (sup-t) band on wave-pair CDF differences from the stratified-PSU bootstrap (Rao and Wu, 1988; Wolter, 2007)—an asymptotic design-consistency guarantee in which country exchangeability plays no role. The conformal *transport* band of Theorems 3–5 calibrates cross-country prediction and the regime diagnostics. Headline counts below are certification counts; the theory sections' finite-sample language does not transfer to them.

The counts collapse up the hierarchy. For `trstprl` over ESS rounds 9–11 (fieldwork 2018–2024), a marginal pairwise reading flags 20 of 30 countries; requiring an any-pair design-aware band drops this to 12; the net first-to-last decline holds in 6 (Austria, Belgium, Estonia, the United Kingdom, Greece, the Netherlands); and the *persistent, design-aware simultaneous* band over the preregistered low-trust core of the scale—the strongest rung—is certified in **one** country, Greece, which also survives the Bonferroni correction across countries. One disclosure is essential to reading that count: *Greece fielded only rounds 10 and 11 in this window*, so its trajectory is a single wave pair and the persistent, net, and any-pair rungs coincide for it—the top rung degenerates, for the sole surviving country, to the strongest available version of a one-pair claim (the same holds for Czechia and Israel, which do not certify). Among the 27 countries with two pairs, *none* certifies persistence, while the

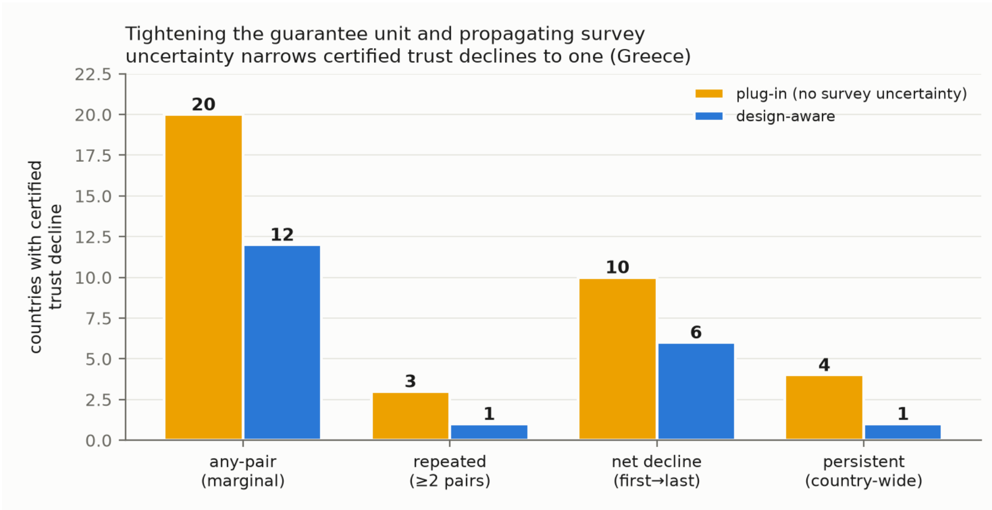

Figure 3: Certified declining countries collapse up the guarantee hierarchy (pairwise → any-pair → net → persistent design-aware; ESS rounds 9–11, $K = 30$ countries, $\alpha = 0.10$), from 20 marginally to 6 net to one persistent (Greece, a single-pair trajectory).

United Kingdom, Belgium, and the Netherlands pass the *plug-in* persistent rung and are demoted by the design-aware layer alone. The honest co-headline is therefore: **net decline in six of thirty; persistent decline in one, and that one on a single pair**. The severity analysis (Section 6) explains why the net rung must carry the secular-decline question: the persistent rung has 80% power only against crisis-scale declines, so its non-certifications are weak evidence of absence, while a certification that passes it is correspondingly strong. For `stfdem` the pattern repeats (23 marginal, 14 any-pair, 8 net, one persistent—Greece, again one pair), though there Greece does *not* survive the Bonferroni rung. Greece is the case the literature regards as a crisis-era legitimacy collapse (Armingeon and Guthmann, 2014; Torcal, 2014), though those accounts concern 2009–2015 and the certified pair is 2020/21→2023/24, a later episode.

The reclassified countries, and the honest word for them. They are the weak-signal, higher-noise cases (scatter in the Supplementary Material): *certified by the marginal plug-in but inconclusive under the design-aware simultaneous band*—never "false positives," a term we reserve for simulation with known truth. Because understated design variance would make *more* countries certify (the m-of-m bias, Section 9), the counts err inclusive, protecting the collapse-of-counts reading though not any individual certification.

Two confounds this window cannot exclude. Round-10 fieldwork coincided with the pandemic's documented rally-round-the-flag trust movements, so "persistent decline through this window" is a claim made against a rally; and several countries switched to self-completion modes at round 10, confounding wave-pair CDF shifts with mode effects—a bias problem no design bootstrap absorbs (Section 9). A mode/fieldwork audit and mode-constant reruns require the microdata and are flagged for the archived replication. Both confounds cut against strong readings in either direction—one more reason the net rung carries the substantive weight of this window.

Robustness across age groups. A preregistered age-group analysis confirms "only Greece" is not an artifact of pooling ages (for trust in parliament it holds within both the 50+ and 30–49 groups; for satisfaction with democracy within 50+); youth certify zero persistent declines as an underpowered null, not youth stability (Supplementary Material).

A second, global target: Foa–Mounk deconsolidation. We preregistered a reanalysis of the "democratic deconsolidation" thesis (Foa and Mounk, 2016, 2017) on the WVS/EVS trends (1981–2022, 59–77 countries with ≥ 2 waves), recoding the five-item battery so that deconsolidation is a persistent decline over each scale's preregistered core. The verdict is calibration, not denial. A *marginal wave-pair reading of the same data*—a criterion we construct to instantiate the bottom rung, not one any published author uses; Foa and Mounk's own empirics are cohort-level comparisons—flags 36–64 countries per item; the persistent, weights-aware object is certified in 6–17 (8–24%), a 2.6–6.5× gap between rungs, with the design-aware layer demoting a further 17–40% per item. Three caveats govern these counts. The WVS ships no PSU/stratum identifiers, so the weights-only bootstrap understates design variance under clustered designs; understated variance can only *inflate* certified counts, so the rung gap is, if anything, a lower bound (a deff-sensitivity rerun is flagged for the archived replication). The per-item lists carry no across-country familywise control—at $\alpha = 0.10$ over 59–77 screened countries a handful of boundary-null certifications per item is expected—which is why we aggregate to multi-item co-certification before interpreting geography. And the certification sets ($K = 59$–77, the ≥ 2-wave subsets) sit below the $K \geq 94$ bound, so every WVS band uses the full-α anchors of Theorem 5; Section 7's $K \approx 95$–105 gate probe describes the full item-coverage sets, a different denominator. The youth "embrace authoritarianism" claim is only half-supported (Supplementary Material).

The certified core: where the surviving deconsolidation lives. Aggregating per-item certifications yields a positive characterization (Figure 4; E30): of 38 countries certified on ≥ 1 item, 13 form a *certified core* on ≥ 2—Albania on four; Bosnia, Ecuador, Tunisia on three; Azerbaijan, Finland, Ghana, Iraq, Lebanon, Rwanda, Switzerland, Trinidad and Tobago, Uzbekistan on two. Three features bear on the debate, each with a stated caveat. *First, the core is not the consolidated West.* It is dominated by post-communist states and the Arab-Spring aftermath, with fragile Latin American and African cases alongside—not the mature democracies the thesis (Foa and Mounk, 2016, 2017) was formulated about. This partly formalizes, with per-country error control, what the post-communist disillusionment literature described qualitatively—decline from euphoric transition-era baselines (Mishler and Rose, 1997; Inglehart and Catterberg, 2003)—and several core members (Rwanda, Uzbekistan, Azerbaijan) are electoral autocracies, where falling stated support reflects autocratic legitimation and possible preference falsification, not "deconsolidation" in the consolidated-democracy sense. The West enters the *core* exactly twice (Finland, Switzerland, on the strongman/army pair), with single-item entries elsewhere: Canada and Sweden on importance of democracy, Norway and the United Kingdom on parliament confidence. The accurate summary: no consolidated Western democracy certifies on more than one item, and none on support for a democratic system—consistent with Wuttke et al. (2022)—not that the West is absent. *Second, the Finnish–Swiss pair is flagged, not headlined.* Persistent rising openness to strongman/army rule in two of the most stable democracies contradicts most published accounts; precisely these items carry documented translation and response-option instabilities, and both countries' trend files interleave EVS and WVS administrations—a distribution-wide shift is exactly what an administration change produces, and our machinery corrects sampling error only. We report the pair as a candidate finding conditional on a within-program measurement audit (microdata required). *Third, the celebrated backsliding cases are narrower than their reputation—with a window caveat.* Turkey, the Philippines, and Mexico certify only on importance of democracy, an item entering the battery only from the mid-2000s; "only on imp_dem" therefore mixes substance with differential window length, and the reading—persistent downgrading of democracy's importance without wholesale attitudinal collapse—should be held loosely. Wave coverage varies by country and item, so multi-item certification is harder where fewer are measured (a per-country wave/program table is flagged for the archived replication). The counts are descriptive aggregation; the regional labels are groupings, not covariates.

9 Discussion

Uncertainty in claims about repeated cross-national surveys is usually computed for the wrong unit—twice. We gave a multiplicity-honest simultaneous band for the claim unit and a hard boundary for the calibration unit: non-identified without the design-noise law (Theorem 1), unreachable at cross-national scale with it (Proposition 1). The honest deployed object is the reduction—a clustered band exact at any K inside a selection-free selector (Theorems 3, 5)—and the design-aware layer certifies, rather than corrects, at survey scale.

When it reduces to ordinary conformal. On real cross-national data the method reduces *exactly* to clustered population conformal (Section 7): the machinery certifies that the simple band is the honest one and reports the ρ that justifies it. The deconvolution regime needs both many exchangeable units and comparable design noise—never both at survey scale, but routinely in many-unit clustered settings. A simulation of that setting (E31, Supplementary Material) confirms the payoff is real: at $K \approx 220$–300 small areas with posterior noise ≈ 0.7 of the between-area signal, the frozen pipeline activates deconvolution on up to 98% of draws at 0.69–0.74× the conservative width with valid coverage—the positive-regime frontier the survey-scale impossibility leaves open.

Limitations. Four scope conditions, stated fully in the Supplementary Material. *Sampling error only:* the machinery is silent on measurement non-invariance (wording and translation changes, EVS/WVS house effects, ESS round-10 mode switches), which produces exactly the shifts the bands certify; certifications straddling such changes are conditional on measurement constancy. *Variance-estimator bias:* the deployed m-of-m PSU bootstrap (Rao et al., 1992; Wolter, 2007) biases design variance downward; a rescaled variant now ships, and reruns are flagged for the archived replication. *A superpopulation postulate:* transport-band exchangeability is over a non-sampled, panel-attriting country set (the certifications do not use it). *Metadata:* without design metadata (WVS) only weights-only replication is possible, and the Gaussian-calibrated finite-K remainder is not fully robust under strong dependence—confined by Theorem 5 to the $K \geq 94$ regime.

The general lesson. Attaching uncertainty to the right unit changes substantive conclusions: net decline in six of thirty European democracies with persistence certifiable in one; a several-fold over-count of persistent deconsolidation, with the surviving core concentrated outside the consolidated West. The pattern is not confined to surveys: wherever a conformal or prediction procedure is calibrated on estimated objects—small-area model outputs, learned features, meta-analytic effects (Remark 1)—the calibration targets carry their own error, and the claim unit is often coarser than the trend asserted. The right response is not to widen every band reflexively but to state the strongest object the data support, correct the estimation noise when it can be identified, and abstain honestly when it cannot.

A Proofs and efficiency results

Full statements and proofs of all theorems, the efficiency results, and the modulation self-inclusion deficit are in Section S1 of the supplementary material; each theorem is paired with a machine-checked contract test.

B Frozen selector specification

The frozen constants, their development-grid derivation, and the validation protocols are in the replication package.

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Competing Interests

Competing interests: the author(s) declare none.

Data and Code Availability

All simulation, theory, and benchmark results are deterministic, reproducing exactly under fixed seeds with no external data, and each theorem is paired with a machine-checked contract test. The ESS, World Values Survey, and AmericasBarometer/LAPOP microdata are licensed to their providers and not redistributed; the replication package documents the exact variables, waves, and retrieval steps. The full package—the `pcb` module, all simulations and contract tests, and the analysis pipeline—will be deposited in the Political Analysis Dataverse (Harvard Dataverse) upon conditional acceptance.

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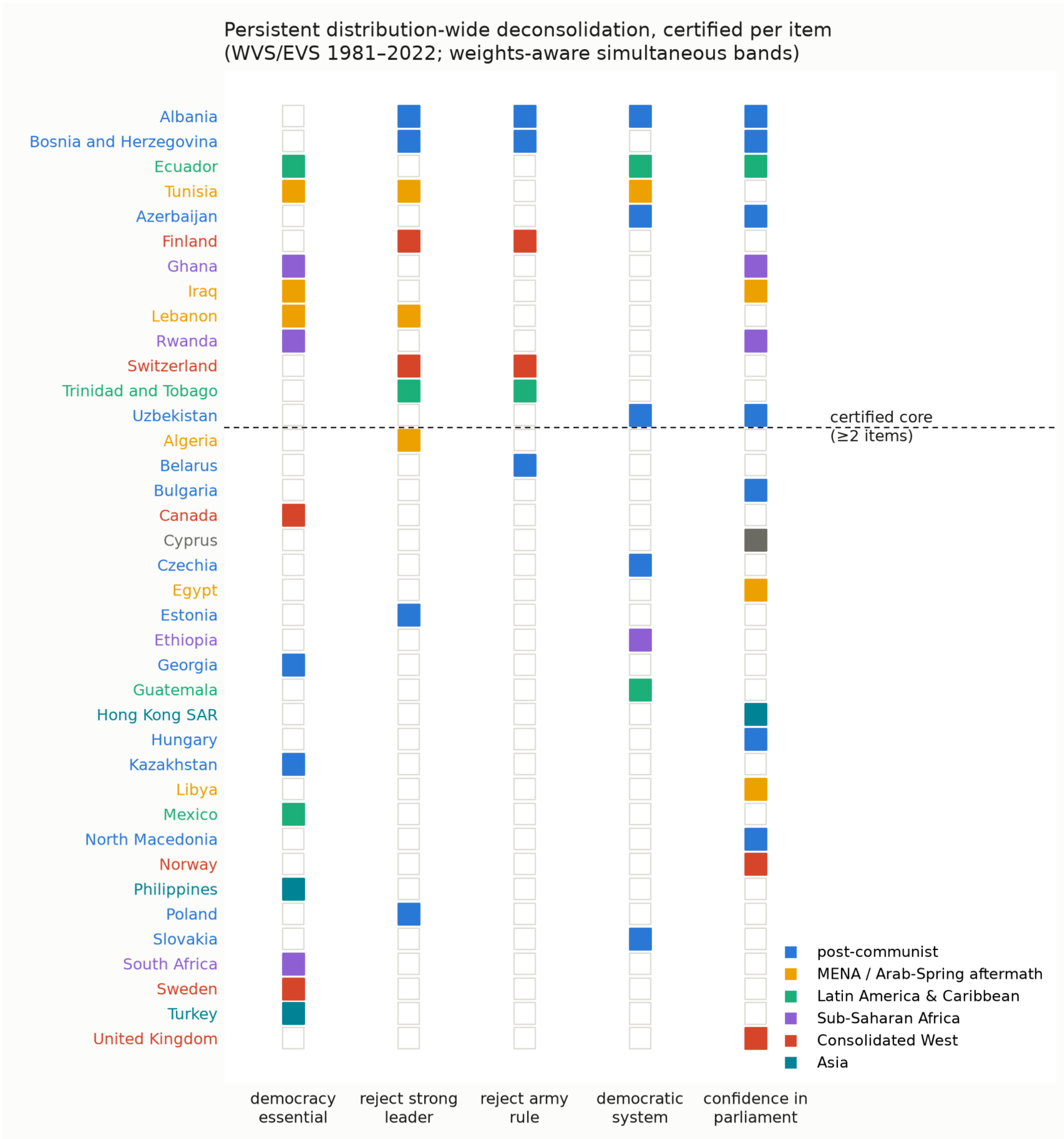

Figure 4: Co-certification across the five battery items (WVS/EVS 1981–2022; $K = 59$–77 per item, $\alpha = 0.10$). Filled cells: persistent weights-aware decline over the preregistered core; countries above the dashed line form the ≥ 2-item certified core.